In [3]:
```
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score,confusion_matrix
```

In [4]:
```
df=pd.read_csv("car_acceptability.csv")
df
```

Out[4]:

	buying price	maintenance cost	number of doors	number of persons	lug_boot	safety	evaluation
0	vhigh	vhigh	2	2	small	med	unacc
1	vhigh	vhigh	2	2	small	high	unacc
2	vhigh	vhigh	2	2	med	low	unacc
3	vhigh	vhigh	2	2	med	med	unacc
4	vhigh	vhigh	2	2	med	high	unacc
...	...	...	...	...	...	...	...
1722	low	low	5	5	med	med	good
1723	low	low	5	5	med	high	vgood
1724	low	low	5	5	big	low	unacc
1725	low	low	5	5	big	med	good
1726	low	low	5	5	big	high	vgood

1727 rows × 7 columns

In [8]:
```
le = LabelEncoder()

df["buying price"]=le.fit_transform(df["buying price"])
df["maintenance cost"]=le.fit_transform(df["maintenance cost"])
df["lug_boot"]=le.fit_transform(df["lug_boot"])
df["safety"]=le.fit_transform(df["safety"])
df["evaluation"]=le.fit_transform(df["evaluation"])
```

In [9]:
```
df
```

Out[9]:

	buying price	maintenance cost	number of doors	number of persons	lug_boot	safety	evaluation
0	3	3	2	2	2	2	2
1	3	3	2	2	2	0	2
2	3	3	2	2	1	1	2
3	3	3	2	2	1	2	2
4	3	3	2	2	1	0	2
...	...	...	...	...	...	...	...
1722	1	1	5	5	1	2	1
1723	1	1	5	5	1	0	3
1724	1	1	5	5	0	1	2
1725	1	1	5	5	0	2	1
1726	1	1	5	5	0	0	3

1727 rows × 7 columns

In [11]:
```
X= df.drop("evaluation",axis=1)
Y=df["evaluation"]
```

In [12]:
```
X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.2,random_state=42
```

In [13]:
```
rf=RandomForestClassifier(n_estimators=100,random_state=42)
rf.fit(X_train,Y_train)
```

Out[13]:

▾ RandomForestClassifier

▶ Parameters

In [14]:
```
Y_pred=rf.predict(X_test)
```

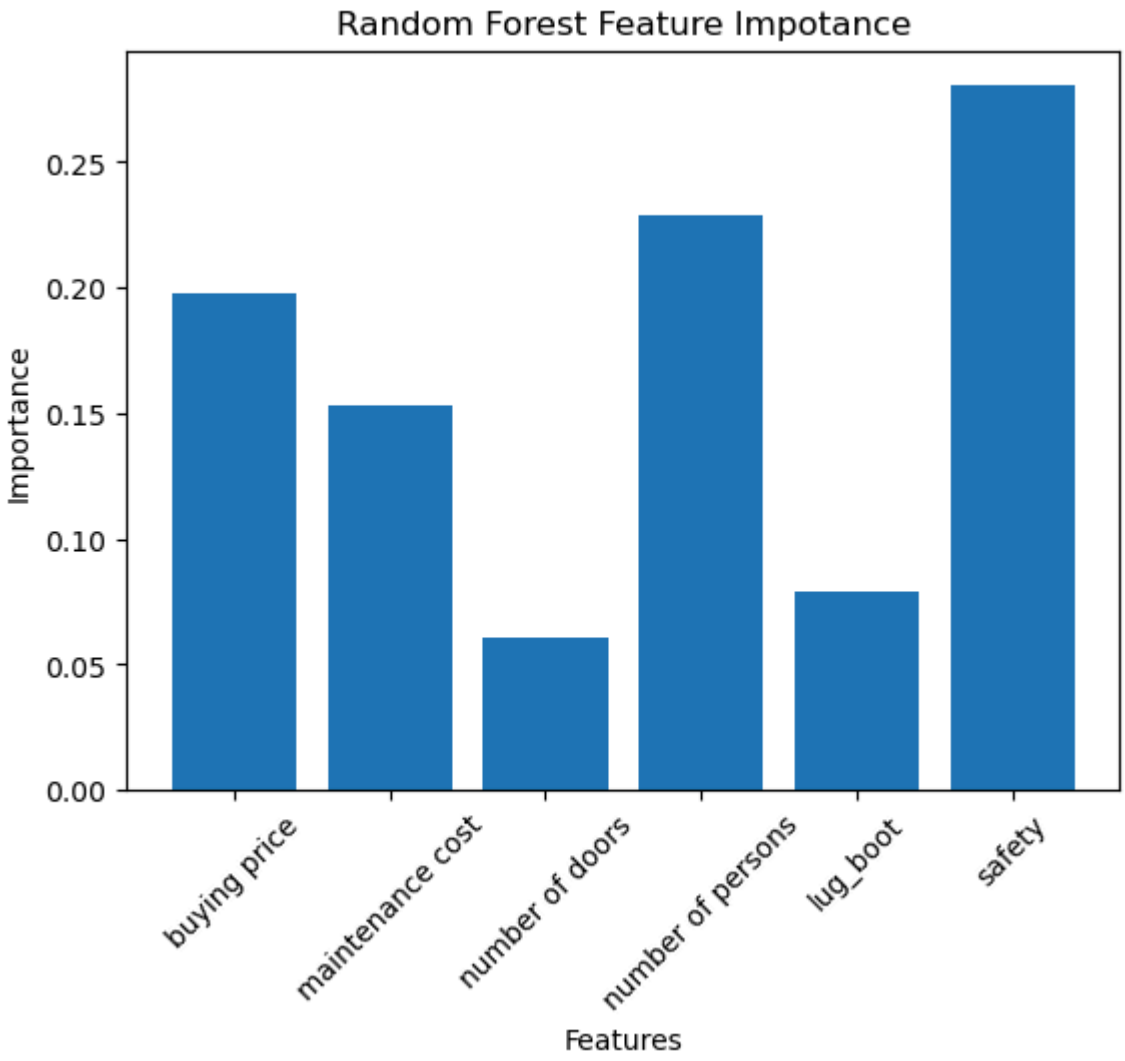
In [16]:
```
print("Accuracy:",accuracy_score(Y_test,Y_pred))
print("\nConfusion Matrix:\n",confusion_matrix(Y_test,Y_pred))
```

```
Accuracy: 0.9624277456647399

Confusion Matrix:
 [[ 72   1   3   1]
 [  2  10   0   3]
 [  1   0 236   0]
 [  2   0   0  15]]
```

In [19]:
```
import matplotlib.pyplot as plt
importance = rf.feature_importances_
plt.bar(X.columns,importance)
plt.xticks(rotation=45)
plt.xlabel("Features")
plt.ylabel("Importance")
```

```
plt.title("Random Forest Feature Impotance")
plt.show()
```



In [ ]: